Project Portfolio · Magic.fit · 2026

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ForMate

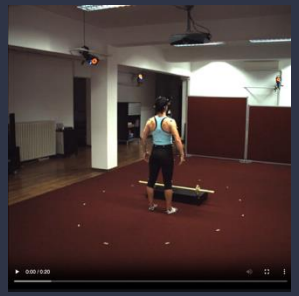
AI-Powered Exercise Form Analysis

Full-stack computer vision pipeline: Pose extraction → Rep segmentation→ Biomechanical scoring → LLM coaching report

3-Stage	100%	Reps	Evaluated
ML Pipeline	Confidence	Auto-detected	Form Score

SYSTEM ARCHITECTURE

Video Input



- The user can choose between exercises
- User can choose between camera angles front oblique or side

01 BRONZE
Pose Detection

- MediaPipe Pose model (33 landmarks)
- Frame-by-frame JSON serialisation
- Normalised coordinates (0–1 per axis)
- OpenCV skeleton overlay + video render

02 SILVER
Rep Segmentation

- Hip/shoulder angle time-series
- Phase detection: concentric / eccentric
- Peak detection + rep boundary ID
- Outlier removal via IQR filtering

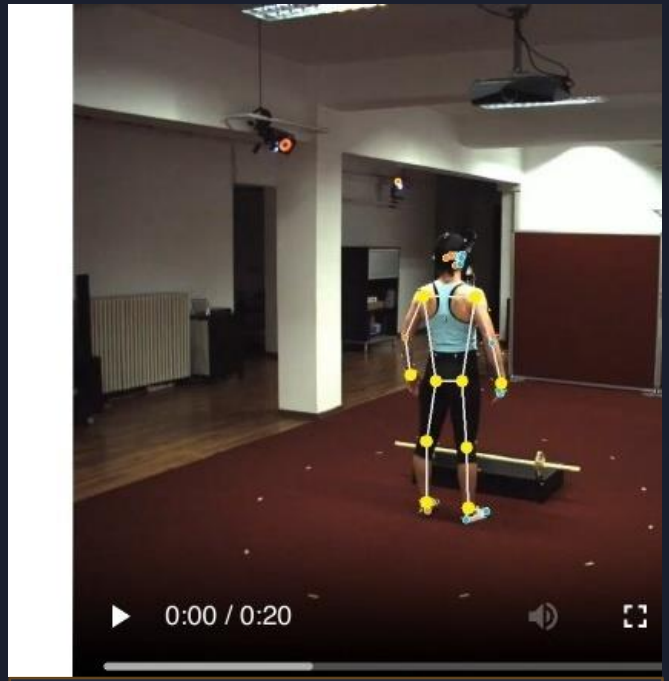
03 GOLD
Metrics & Coaching

- 5-dimensional form scoring engine
- Trunk instability + tempo CV flags
- Snapshot extraction at flagged frames
- LLM coaching report (Claude API)

OUTPUT

Python · MediaPipe · OpenCV · NumPy / Pandas · Streamlit · Claude API (LLM)

STAGE 01 — BRONZE: POSE DETECTION & OVERLAY



MediaPipe skeleton overlay — 33 keypoints tracked per frame

Model

Google MediaPipe Pose (BlazePose backbone)
33 landmark keypoints per frame including face, body, hands and feet

Coordinate System

Normalised x/y/z per keypoint (0.0–1.0 relative to frame)
Visibility confidence score also stored per landmark

Output Format

JSON per frame: { frame_id, landmarks: [{x,y,z,vis}×33] }
VideoWriter renders skeleton overlay back to MP4

Configuration

Exercise type: deadlift · Camera: front_oblique
Min detection confidence: 0.5 · Min tracking: 0.5

Session: 20260303_102844_deadlift · Video: deadlift_2.mp4 (7.3 MB, 20s) · Bronze output: pose_data.json + overlay.mp4

STAGE 02 — SILVER: REP SEGMENTATION & CLEANING

Processing Pipeline

1. **Angle Extraction**
   Compute hip/shoulder angle θ per frame from normalised landmark coords (arctan2)

2. **Signal Smoothing**
   Rolling mean window applied to angle time-series to reduce noise from jitter

3. **Peak Detection**
   scipy.signal.find_peaks on inverted angle series → rep boundary frames identified

4. **Phase Labelling**
   Frames between boundaries labelled concentric (pull) or eccentric (descent)

5. **Outlier Removal**
   IQR filter on rep duration removes anomalous reps (wrong movement, camera occlusion)

Per-Rep Output Metrics

Rep	t_pull (s)	t_descent (s)	Ratio
1	0.94	1.18	0.80×
2	0.50	1.48	0.34×
3	0.54	1.64	0.33×
4	0.54	1.72	0.31×
5	0.60	1.74	0.34×
6	0.54	2.32	0.23×

⚠ Tempo Flag Triggered

Pull time CV = 22.8% → exceeds threshold (15%). Coefficient of variation computed across 6 reps — flag passed to Gold stage.

Silver output: rep_segments.json · 6 valid reps extracted · 0 reps removed by IQR filter

STAGE 03 — GOLD: SCORING ENGINE + LLM COACHING

OVERALL SCORE

51.1

/100

score_output.json

```
{
 "setup": 100,
 "hinge": 100,
 "symm":  17.2,
 "trunk":  0,
 "tempo":  17.8,
 "overall": 51.1
}
```

$$\text{overall} = \Sigma(w_i \times \text{score}_i)$$
where $\Sigma w_i = 1$

Scoring Formula per Dimension

Setup Consistency — 100%

= 1 if std(hip_x) < 0.05 else 0

Hinge Quality — 100%

= mean(hip_angle > 45°)

Symmetry — 17%

= 1 − |left_hip_y − right_hip_y|

Trunk Control — 0%

= 1 if std(trunk_angle) < 8° else 0

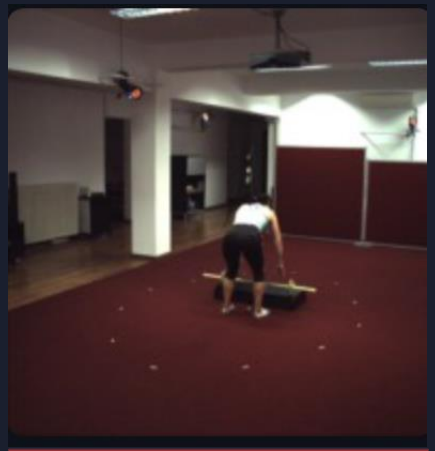
Tempo Consistency — 18%

= 1 − CV(t_pull) across reps

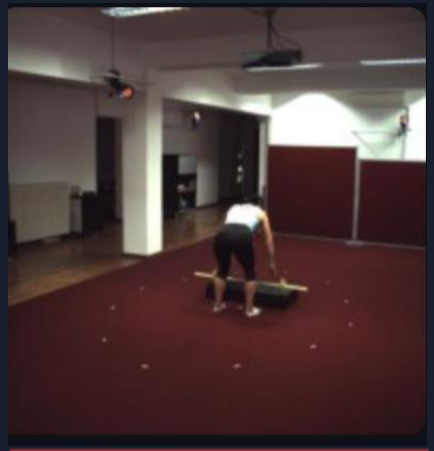
Gold → Claude API: per-rep metrics + flags injected into structured prompt → personalised coaching Markdown report

FORM ISSUE SNAPSHOTS — LIVE SESSION OUTPUT

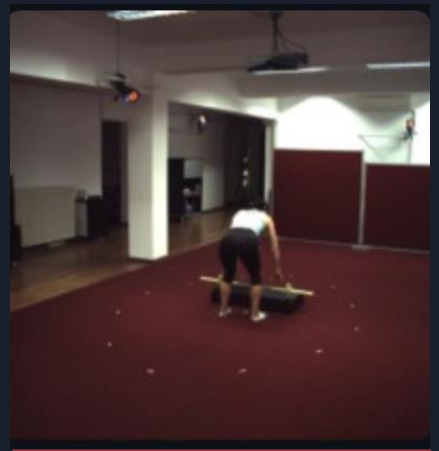
Frames automatically extracted where torso angle exceeds 25° threshold — flagged by Gold scoring engine as trunk instability events.



Frame 108 — Trunk instability detected



Frame 109 — Trunk instability detected



Frame 110 — Trunk instability detected

Snapshot Extraction Logic

```
for frame in rep_frames:
  trunk_angle = calc_trunk_angle(landmarks[frame])
  if trunk_angle > THRESHOLD (25°): → extract_snapshot(frame)
```

Why ForMate?

Technical skills demonstrated

🧠 ML / CV Pipeline

Pose estimation, temporal signal processing, landmark normalisation, peak detection

⚙️ Data Engineering

Medallion architecture (Bronze/Silver/Gold), JSON serialisation, IQR filtering, metric extraction

🤖 AI / LLM Integration

Claude API prompt engineering, structured output generation, context injection from computed metrics

💻 Software Engineering

End-to-end Python app, modular pipeline, Streamlit UI, OpenCV video I/O, Git version control

🔬 Domain Translation

Biomechanics → computable metrics. Physical science principles encoded as algorithmic checks

🎯 Product Thinking

Minimal-setup UX, actionable scoring, automated coaching — designed for real end-users